Supplementary material for the work "Electron-Ion Path Integral Monte Carlo"

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Supplementary material to the article "Electron-Ion Path Integral Monte Carlo".

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I. INTRODUCTION

We here report the FORTRAN code listing of the code used in the simulations developed in the publication "Electron-Ion Path Integral Monte Carlo". This is shown in Appendix A. The model studied in that publication was that of a 1:1 electron-proton plasma. A particular Two Component Plasma (TCP) where the two species are both fermions, one about 2000 times more heavy than the other. A particular neutral binary mixture with Coulomb pair interactions.

The Restricted Path Integral Monte Carlo (RPIMC) method used in our computer experiments is the one described in Ref. [1, 2]. But the permutation sampling necessary in calculating the determinant required by the study of the Fermi-Dirac statistics was carried out proposing swaps of a randomly chosen pair of particles through the Lévy construction of a pair of Brownian bridges (see Ref. [3] or Section V.G of Ref. [1] and references therein) connecting the two paths.

In the simulation we measure some thermodynamic properties like the total kinetic and potential energy and the superfluid fraction of each species. And some structural properties like the partial Radial Distribution Functions (RDF) for the two species.

The plane waves [1] Path Integral Monte Carlo (PIMC) method used in our computer experiments is the one described in Ref. [1]. A single MC step is made of a swap of a randomly chosen pair of particles through the *Lévy construction* of two Brownian bridge between two randomly chosen timeslices on the two different paths (see Section V.G of Ref. [1] and references therein for the multislice move) and of a displacement of all the timeslices of a particle path chosen at random (the singleslice move). This allows the permutation sampling necessary in calculating the symmetric density matrix required by the study of the Bose-Einstein statistics. We then introduced a path restriction in order to overcome the "fermions sign problem" [5] based on the sign of the free fermions density matrix calculated in subroutine `ifdm`. This results in an approximate RPIMC calculation [2] which calculates the antisymmetric density matrix required by the study of the Fermi-Dirac statistics. Of course we only apply the swap to like particles of the same species.

The extent of the singleslice displacement is chosen at the beginning of the run through the input variable `ALPHA` or it can be adjusted during the run every `IRATIO` MC steps so to have acceptance ratios closest to 1/2 (this functionality is commented out in the present listing).

We discretize the imaginary time between 0 and `BETA` into `FTNO` timeslices of length `LS=BETA/FTNO`. One usually wants to compare simulations at fixed `LS`. The multislice move requires the reconstruction of a randomly chosen number `MBM≤MBMM≤FNTO` of timeslices between a randomly chosen initial timeslice `CP` on a randomly chosen particle `IP` and the final timeslice on a randomly chosen particle `KP`. Here `MBMM` is the maximum number of timeslices in a Brownian bridge. The singleslice move requires the displacement of all the `FTNO` timeslices of the path of the particle `IP` chosen at random. So a single MC step requires a maximum of `2MBMM+FTNO` timeslice moves. In the code we wrote the FORTRAN instructions for the multislice move in lowercase and the ones for the singleslice move in uppercase. The simulation uses only the multislice move if the logical variable `IFBRIDGE` is `TRUE` and only the singleslice move if the logical variable `IFDISP` is `TRUE`. The simulation lasts for a total number of MC steps `NSTEP`. All these variables can be chosen in the `data-mix.in` simulation configuration input file read at startup.

The code works in any space dimension `DIM≤MDIM` with a mixture with a number of components `NMIX≤MMIX`. The number of particles of species `I` is `NPM(I)` and they have mass `FTM(I)`. One has to specify also the thermodynamic configuration giving the density `DENS` and the absolute temperature `TEMP`.

The pair potential between the particles `POTK` is encoded in `SUBROUTINE POT` at the end of the code listing. It includes the AQHS effective potential (`QHS`) and the canonical HS (`BUMP`). Other test cases included are the harmonic potenial (`HARMONIC`), the Coulomb potential (`COULOMB`), the penetrable square well potential (`PSW`), and the Lennard-Jones potential (`LJ`). The ideal gas is obtained choosing the string `FREE` in the `data-mix.in` simulation configuration input file. The potential can be further specified through the variables `RCUT`, `SIG`, `EPSR`, `EPSA`, and `EPS` in the simulation configuration input file. For the sake of the present work we only use the Coulomb potential where `SIG(I,J)` is given by Eq. (2.8) in the

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[1] For a coherent states PIMC see Ref. [4].

main text of the article and `EPSA(I,J)` is given by Eq. (2.9) in the main text of the article.

The simulation calculates both thermodynamic quantities and static structural properties. For the thermodynamic quantities we measure: the kinetic energy `KE`, the potential energy `V`, and the superfluid fraction of each species `SF(I)`. These quantities are stored in the output file `fort.9` every `IPRINT` MC steps. Whereas the path configurations output file `CNFILE` is stored in a $.xyz$ format every `ISAVE` MC steps. The first `IEQUI` MC steps are used for the equilibration of the Markov chain. Quantities are only calculated after these equilibration steps. In order to determine efficiently the superfluid fraction at low temperature, which requires large `FTN0`, it is convenient to deactivate the displace move choosing `TRUE` for `IFBRIDGE`. For the structural properties we measure the partial radial distribution function $g_{ij}(r)$ which are written in the output files `rdf11.dat`, `rdf12.dat`, `rdf22.dat`. In the current implementation of the code, the $g_{ij}(r)$ are only calculated during the displacement move every `ICALCG` timeslices.

The logical variable `IFB` is set in the simulation configuration input file. If it is set to `TRUE` permutations of particles are sampled, otherwise only Boltzmann statistics for distinguishable particles is used.

If the variable `INIT` is set to 0 it starts from a path with the particles randomly distributed uniformly with no overlaps with all timeslices copied above the first one for each particle, if it is set to 1 it reads the paths configuration from the file `CNFILE` stored on disk, if it is set to 2 it does a complete restart from the file `CNFILE` and the frozen configuration `fort.20` both stored on disk every `ISAVE` MC steps.

The variable `SEED` is also set in the simulation configuration input file `data-mix.in`. It is the positive integer seed of the pseudo random number generator.

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Appendix A: The code

This is the code used for the PIMC computer experiment. We list here the main FORTRAN code `rpimc-mix.f` with its included `mc-mix.par` parameters file. And the input data file `data-mix.in` (for cases G and H of the Table in the main text of the article) that is read at the beginning of the run.

```
*****************************************************************************
***  rpimc-mix.f  ***********************************************************
*****************************************************************************

        PROGRAM RPIMC
        IMPLICIT NONE
C     ***************************************************************
C     ** MONTE CARLO SIMULATION PROGRAM FOR [PW] RESTRICTED PATH      **
C     ** INTEGRAL FOR A BINARY MIXTURE                                **
C     **                                                              **
C     ** space dimensions:    DIM=1,2,3                               **
C     ** number of particles: NP                                      **
C     ** number of timesteps: FTN0                                    **
C     ** units:               hbar=k_B=1                              **
C     ** BETA=1/TEMP=FTN0*LS  (LS imaginary time spacing)             **
C     **                                                              **
C     ** OBSERVABLES:                                                 **
C     ** kinetic energy       KE                                      **
C     ** potential energy     V                                       **
C     ** forces virial        W                                       **
C     ** superfluid fraction  SF                                      **
C     ** radial distribution function in                              **
C     ** 'rdf11.dat', 'rdf12.dat', 'rdf22.dat'                        **
C     ******************************************************************
        INCLUDE       'mc-mix.par'                          ! parameters

        INTEGER*4     SEED
        INTEGER*8     IDIM, STEP, I, J, K
        INTEGER*8     C1, CP, IP, KP, NIP, NKP, LL, OUT1, OUT2, PIP
        INTEGER*8     INIT, NSTEP, NS, IPRINT, ISAVE, IRATIO, IEQUI
        INTEGER*8     MBMM, MCM, PREV(MNP)
        INTEGER*8     RS

        REAL*8        DENS, TEMP, BETA, DENSLJ
        REAL*8        DRMAX, ALPHA, CDIM
        REAL*8        RANF, DUMMY, SR9, SR3
        REAL*8        VLRC, VLRC6, VLRC12, WLRC, WLRC6, WLRC12
        REAL*8        ACM, ACATMA, ACATMAS, ACATMAB
        REAL*8        V, VNEW, VOLD, VEND, DELTV, VN, VS
        REAL*8        W, WNEW, WOLD, WEND, DELTW, WN, WS
        REAL*8        KE, KENEW, KEOLD, DELTKE, KEEND
        REAL*8        AVV, ACV, ACVSQ, FLV
        REAL*8        AVW, ACW, ACWSQ, FLW
        REAL*8        AVKE, ACKE, ACKESQ, FLKE
        REAL*8        ACT, ACTNEW, ACTOLD, DELTACT, DELTACTB
        REAL*8        RXIOLD(MDIM), RXINEW(MDIM)
        REAL*8        RXP(MDIM,0:N), RXPP(MDIM,0:N)
        REAL*8        STD(MMIX), PS, KKK, VVV, WWW
        REAL*8        RATIO, RATIOS, RATIOB
        REAL*8        WIND(MMIX,MDIM), ACWINDSQ(MMIX,MDIM)
        REAL*8        SFI(MMIX), SF(MMIX), WSP(MMIX)
        REAL*8        IFDMS, RXOLD(MDIM,0:N,MNP)

        INTEGER*8     BIN, MAXBIN, MAXBIN2, ICALCG
        INTEGER*8     HIST11(MBIN),HIST12(MBIN),HIST22(MBIN)
        REAL*8        CONST, RLOWER, RUPPER, TSTEP
        REAL*8        NIDEAL11, NIDEAL12, NIDEAL22
        REAL*8        GR11, GR12, GR22, CO(MMIX)
        REAL*8        CONST11,CONST12,CONST21,CONST22

        CHARACTER     CNFILE*30, POTK*10

        LOGICAL       OVRLAP, IFB, IFDISP, IFBRIDGE

C        PI = ACOS(-1.d0)
        OPEN (UNIT = 11, FILE = 'rdf11.dat', STATUS ='UNKNOWN')
        OPEN (UNIT = 12, FILE = 'rdf12.dat', STATUS ='UNKNOWN')
        OPEN (UNIT = 13, FILE = 'rdf22.dat', STATUS ='UNKNOWN')
        OPEN (UNIT = 14, FILE = 'rdfto.dat', STATUS ='UNKNOWN')
C     ******************************************************************
C     ** READ INPUT DATA                                              **
C     ******************************************************************

        WRITE(*,'('' ********** PROGRAM PIMC ************      ''/)')
        WRITE(*,'('' PLANE WAVES                                  ')')
        WRITE(*,'('' PATH INTEGRAL MONTE CARLO PROGRAM            ')')
        OPEN (UNIT=10, FILE='data-mix.in', STATUS='UNKNOWN')
        READ (10,*) I
        READ (10,*) DIM, NMIX
        READ (10,*) I
        READ (10,*) SEED
        READ (10,*) I
        READ (10,*) IFB
        READ (10,*) I
        READ (10,*) (NPM(J),J=1,NMIX)
        READ (10,*) I
        READ (10,'(A)') POTK
        READ (10,*) I
        READ (10,*) FTN0
        READ (10,*) I
        READ (10,*) (FTM(J),J=1,NMIX)
        READ (10,*) I
        READ (10,*) NSTEP
        READ (10,*) I
        READ (10,*) IPRINT
        READ (10,*) I
        READ (10,*) ISAVE
        READ (10,*) I
        READ (10,*) IEQUI
        READ (10,*) I
        READ (10,*) IRATIO
        READ (10,*) I
        READ (10,*) ICALCG
        READ (10,*) I
        READ (10,'(A)') CNFILE
        READ (10,*) I
        READ (10,*) INIT
        READ (10,*) I
        READ (10,*) DENS
        READ (10,*) I
        READ (10,*) TEMP
        READ (10,*) I
        READ (10,*) ALPHA
        READ (10,*) I
        READ (10,*) MBMM
        READ (10,*) I
        READ (10,*) RCUT
        READ (10,*) I
        READ (10,*) ((SIG(I, J),I=1,NMIX),J=1,NMIX)
```

```
      READ (10,*) I
      READ (10,*) EPSR
      READ (10,*) I
      READ (10,*) ((EPSA(I, J),I=1,NMIX),J=1,NMIX)
      READ (10,*) I
      READ (10,*) EPS
      READ (10,*) I
      READ (10,*) IFDISP
      READ (10,*) I
      READ (10,*) IFBRIDGE
      CLOSE (UNIT=10)

      WRITE(*,'('' IN DIMENSION (1,2,3,...)''I12/)') DIM
      WRITE(*,'('' ************************************      ''/)')

      GOTO 1111                    ! comment if input from keyboard

      WRITE(*,'('' ********** PROGRAM PIMC ************      ''/)')
      WRITE(*,'('' PLANE WAVES                                 ''/)')
      WRITE(*,'('' PATH INTEGRAL MONTE CARLO PROGRAM           ''/)')

      WRITE(*,'('' ENTER THE SPATIAL AND MIXTURE DIMENSIONS   ''/)')
      READ (*,*) DIM, NMIX
      WRITE(*,'('' ENTER SEED FOR RANDOM SEQUENCE             ''/)')
      READ (*,*) SEED
      WRITE(*,'('' IF BOSE (T/F)                              ''/)')
      READ (*,*) IFB
      WRITE(*,'('' ENTER THE NUMBER OF PARTICLES < MNP        ''/)')
      READ (*,*) (NPM(J),J=1,NMIX)
      WRITE(*,'('' ENTER TYPE OF PAIR POTENTIAL (Many Body)   ''/)')
      READ (*,'(A)') POTK
      WRITE(*,'('' ENTER THE NUMBER OF DISCRETIZATIONS < N    ''/)')
      READ (*,*) FTNO
      WRITE(*,'('' ENTER THE BARE MASS                        ''/)')
      READ (*,*) (FTM(J),J=1,NMIX)
      WRITE(*,'('' ENTER NUMBER OF CYCLES                     ''/)')
      READ (*,*) NSTEP
      WRITE(*,'('' ENTER NUMBER OF STEPS BETWEEN OUTPUT LINES ''/)')
      READ (*,*) IPRINT
      WRITE(*,'('' ENTER NUMBER OF STEPS BETWEEN DATA SAVES   ''/)')
      READ (*,*) ISAVE
      WRITE(*,'('' ENTER NUMBER OF STEPS FOR EQUILIBRATION    ''/)')
      READ (*,*) IEQUI
      WRITE(*,'('' ENTER INTERVAL FOR UPDATE OF MAX. DISPL.   ''/)')
      READ (*,*) IRATIO
      WRITE(*,'('' ENTER INTERVAL FOR RDF CALCULATION         ''/)')
      READ (*,*) ICALCG
      WRITE(*,'('' ENTER THE CONFIGURATION FILE NAME          ''/)')
      READ (*,'(A)') CNFILE
      WRITE(*,'('' ENTER 0 IF INITIALIZATION NEEDED           ''/)')
      READ (*,*) INIT
      WRITE(*,'('' ENTER THE DENSITY                          ''/)')
      READ (*,*) DENS
      WRITE(*,'('' ENTER THE TEMPERATURE                      ''/)')
      READ (*,*) TEMP
      WRITE(*,'('' ENTER MAX. DISPLACEMENT DRMAX/SIGMA        ''/)')
      READ (*,*) ALPHA
      WRITE(*,'('' ENTER MAXIMUM NUMBER OF BRIDGE TIMESLICES  ''/)')
      READ (*,*) MBMM
      WRITE(*,'('' ENTER THE POTENTIAL CUTOFF DISTANCE (~ 2.5)''/)')
      READ (*,*) RCUT
      WRITE(*,'('' ENTER SIG                                  ''/)')
      READ (*,*) ((SIG(I, J),I=1,NMIX),J=1,NMIX)
      WRITE(*,'('' ENTER EPSR                                 ''/)')
      READ (*,*) EPSR
      WRITE(*,'('' ENTER EPSA                                 ''/)')
      READ (*,*) ((EPSA(I, J),I=1,NMIX),J=1,NMIX)
      WRITE(*,'('' ENTER EPS                                  ''/)')
      READ (*,*) EPS
      WRITE(*,'('' IF DISPLACEMENT MOVE (T/F)                 ''/)')
      READ (*,*) IFDISP
      WRITE(*,'('' IF BRIDGE MOVE (T/F)                       ''/)')
      READ (*,*) IFBRIDGE
      WRITE(*,'('' IN DIMENSION (1,2,3,...)''I12/)') DIM
      WRITE(*,'('' ************************************      ''/)')

 1111 CONTINUE

      IF (MBMM .GT. FTNO) THEN
         WRITE(*,*) "number of timeslices in bridge > ",FTNO
         STOP
      ENDIF

      NP =  0
      DO I=1, NMIX
         NP = NP + NPM(I)
      ENDDO
      DO I=1, NMIX
         CO(I)=DBLE(NPM(I))/DBLE(NP)
      ENDDO

C     ** INITIALIZE RANDOM NUMBER GENERATOR **

      CALL SRAND ( SEED )
      IF (SEED.EQ.1) THEN
         CALL SRAND ( 0 )
      ENDIF

C     ** INVERSE TEMPERATURE BETA **

      BETA = (1.d0/TEMP)

C     ** IMAGINARY TIMESTEP **

      LS  = BETA/FTNO      ! time step
      DO I=1,NMIX
         STD(I) = (LS/FTM(I))**0.5  ! standatd deviation of free-particle rho
      ENDDO

C    ** WRITE INPUT DATA **

      WRITE(*,'('' SEED                        ''',I10   )'') SEED
      WRITE(*,'('' POTENTIAL                   ''',A     )'') POTK
      WRITE(*,'('' DIMENSIONS                  ''',I10   )'') DIM
      WRITE(*,'('' NUMBER OF PARTICLES         ''',I10   )'') NP
      WRITE(*,'('' NUMBER OF CYCLES            ''',I10   )'') NSTEP
      WRITE(*,'('' NUMBER OF EQUIL. STEPS      ''',I10   )'') IEQUI
      WRITE(*,'('' OUTPUT FREQUENCY            ''',I10   )'') IPRINT
      WRITE(*,'('' SAVE FREQUENCY              ''',I10   )'') ISAVE
      WRITE(*,'('' RATIO UPDATE FREQUENCY      ''',I10   )'') IRATIO
      WRITE(*,'('' CONFIGURATION FILE  NAME    ''',A     )'') CNFILE
      WRITE(*,'('' TEMPERATURE                 ''',E10.4 )'') TEMP
      WRITE(*,'('' DENSITY                     ''',E10.4 )'') DENS
      WRITE(*,'('' PARTICLE MASS               ''',E10.4 )'') FTM
      WRITE(*,'('' # TIME SLICES               ''',I10   )'') FTNO
      WRITE(*,'('' TIME STEP                   ''',E10.4 )'') LS

C    ** CONVERT INPUT DATA TO PROGRAM UNITS **

      SIGMA  = ( DBLE ( NP ) / DENS ) ** ( 1.0 / DIM )
      DRMAX  = ALPHA * SIGMA

      IF (POTK .EQ. 'LJ' .AND. NMIX .EQ. 1) THEN
         DENSLJ = DENS * SIG(1,1) ** DBLE( DIM )
         RCUT = SIGMA/2.d0/SIG(1,1)
      ENDIF

      MAXBIN  = INT ( SIGMA/DELR )
      MAXBIN2 = INT ( MAXBIN/2.0 )
      HIST11  = 0.
      HIST12  = 0.
      HIST22  = 0.

      IF (DIM .eq. 3) THEN
         CONST11 = 4.d0 * PI * DENS * CO(1)/ 3.d0
         CONST12 = 4.d0 * PI * DENS * CO(2)/ 3.d0
         CONST21 = 4.d0 * PI * DENS * CO(1)/ 3.d0
         CONST22 = 4.d0 * PI * DENS * CO(2)/ 3.d0
      ENDIF
      IF (DIM .eq. 2) THEN
         CONST11 = PI * DENS * CO(1)
         CONST12 = PI * DENS * CO(2)
         CONST21 = PI * DENS * CO(1)
         CONST22 = PI * DENS * CO(2)
      ENDIF
      IF (DIM .eq. 1) THEN
         CONST11 = 2.d0 * DENS * CO(1)
         CONST12 = 2.d0 * DENS * CO(2)
         CONST21 = 2.d0 * DENS * CO(1)
         CONST22 = 2.d0 * DENS * CO(2)
      ENDIF

C     ** INITIALIZE CONFIGURATION **

      DO I = 1, NP
         NEXT(I) = I
         LEXT(I) = I
         DO J = 1, FTNO
            MEXT(J,I)  = I
         ENDDO
      ENDDO

C     ** ZERO ACCUMULATORS **

      ACV     = 0.0
      ACVSQ   = 0.0
      FLV     = 0.0
      ACW     = 0.0
      ACWSQ   = 0.0
      FLW     = 0.0
      ACKE    = 0.0
      ACKESQ  = 0.0
      FLKE    = 0.0
      ACWINDSQ= 0.0
      ACM     = 0.0
      ACATMA  = 0.0
      ACATMAB = 0.0
      ACATMAS = 0.0

C     ** READ INITIAL CONFIGURATION **
      IF ( INIT .EQ. 0 ) THEN
         PRINT*,"PATHS INITIALIZATION ..............."
         OPEN (UNIT=9, STATUS='UNKNOWN')
         CALL INITCN ( CNFILE ) ! random configuration
         CALL READCN ( CNFILE ) ! random configuration
         NS = NSTEP
      ELSEIF ( INIT .EQ. 1) THEN
         PRINT*,"RESTART FROM CONFIG  ..............."
         OPEN (UNIT=9, STATUS='UNKNOWN')
         CALL READCN ( CNFILE )
         NS = NSTEP
      ELSEIF ( INIT .EQ. 2) THEN
         PRINT*,"RESTART FULL         ..............."
         OPEN (UNIT=9, ACCESS='APPEND')
         CALL READCN ( CNFILE )
         ACATMA  = 0.0
         ACATMAB = 0.0
         ACATMAS = 0.0
         READ(20,*)(HIST11(I),I=1,MAXBIN),
     &              (HIST12(I),I=1,MAXBIN),
     &              (HIST22(I),I=1,MAXBIN),
     &         ACV,ACVSQ,ACW,ACWSQ,ACKE,ACKESQ,ACWINDSQ,ACM,NS
         NS=NS+NSTEP
      ELSE
         WRITE (*,*) 'INIT different from 0,1,2'
         STOP
      ENDIF

C     ** CALCULATE LONG RANGE CORRECTIONS    **
C     ** SPECIFIC TO THE LENNARD JONES FLUID **
      IF (DIM .EQ. 1) CDIM = 1
      IF (DIM .EQ. 2) CDIM = PI
```

```
      IF (DIM .EQ. 3) CDIM = 2*PI

      SR3 = - RCUT ** (- 6. + DIM)/(-6. + DIM)
      SR9 = - RCUT ** (- 12. + DIM)/(-12. + DIM)

      VLRC12 =   4 * EPS * CDIM * DENSLJ * NP * SR9
      VLRC6  = - 4 * EPS * CDIM * DENSLJ * NP * SR3
      VLRC   =   VLRC12 + VLRC6
      WLRC12 =   4.0  * VLRC12
      WLRC6  =   2.0  * VLRC6
      WLRC   =   WLRC12 + WLRC6

C     ** WRITE OUT SOME USEFUL INFORMATION **

      WRITE(*,'('' SIGMA                    '',E10.4)') SIGMA
      WRITE(*,'('' MAXIMUM DISPLACEMENT     '',E10.4)') DRMAX

C    ** CALCULATE INITIAL ENERGY AND CHECK FOR OVERLAPS **

      CALL SUMUP (POTK, OVRLAP, KE, V, W)

      IF (POTK .EQ. 'LJ') THEN
         VS = ( V + VLRC )
         WS = ( W + WLRC )
      ELSE
         VS = V
         WS = W
      ENDIF

      WRITE(*,'('' INITIAL V                '',E10.4)') VS
      WRITE(*,'('' INITIAL W                '',E10.4)') WS
      WRITE(*,'('' INITIAL KE               '',E10.4)') KE

      WRITE(*,'(//'' START OF MARKOV CHAIN              ''//)')
      WRITE(*,'(''  NMOVE     RATIO       ACTION          ''/)')

C     ******************************************************************
C     ** LOOPS OVER ALL CYCLES AND ALL TIME SLICES                    **
C     ******************************************************************

      DO 100 STEP = NS-NSTEP+1, NS

         CP = INT(FTN0*RANF(DUMMY))+1    ! select a random timeslice
C         RS = INT(FTN0*RANF(DUMMY))+1 ! select a random referenceslice
         RS = 1
         MBM = INT((MBMM-1)*RANF(DUMMY))+2 ! # timeslices in bridge
         IP = INT(NP*RANF(DUMMY))+1      ! select a particle
         KP = INT(NP*RANF(DUMMY))+1      ! select another particle
         IF (.NOT.IFB) KP = IP           ! for boltzmann statistics
         IF (IFDISP) GOTO 77   ! uncomment if only displacement move
         IF (ISP(IP).NE.ISP(KP)) GOTO 77 ! different species
C     ******************************************************************
C     ** BRIDGE&SWAP MOVE (BOSE STATISTICS)                           **
C     ******************************************************************
         mcm = cp+mbm-floor((cp+mbm-.1)/ftn0)*ftn0

         rxold=rx

         nip=next(ip)
         nkp=next(kp)
         ps=0.d0

         vold=0.d0
         wold=0.d0
         if(cp+mbm.le.ftn0)then
            do i=cp+1,mcm-1
               call pppnergy ( potk, rx(:,i,ip), ip,
     :              rx(:,i,kp), kp, i,
     :              vvv, www )
               vold=vold+vvv
               wold=wold+www
            enddo
         else
            do i=cp+1,ftn0
               call pppnergy ( potk, rx(:,i,ip), ip,
     :              rx(:,i,kp), kp, i,
     :              vvv, www )
               vold=vold+vvv
               wold=wold+www
            enddo
            do i=1,mcm-1
               call pppnergy ( potk, rx(:,i,nip), nip,
     :              rx(:,i,nkp), nkp, i,
     :              vvv, www )
               vold=vold+vvv
               wold=wold+www
            enddo
         endif

         keold=0.d0
         do k=1,np
            if(k.eq.ip.or.k.eq.kp.or.k.eq.nip.or.k.eq.nkp)then
               do i=1,ftn0
                  call kkknergy ( rx(:,i,k), k, i,
     :                 kkk )
                  keold=keold+kkk
               enddo
            endif
         enddo

         if(cp+mbm.le.ftn0)then
            call bridge(rx(:,cp,ip),rx(:,mcm,kp),std(isp(ip)),
     :           rxp,out1)
            if (ip.ne.kp) then
               call bridge(rx(:,cp,kp),rx(:,mcm,ip),std(isp(ip)),
     :              rxpp,out2)
            endif
         else
            call bridge(rx(:,cp,ip),rx(:,mcm,nkp),std(isp(ip)),
     :           rxp,out1)
            if (ip.ne.kp) then
               call bridge(rx(:,cp,kp),rx(:,mcm,nip),std(isp(ip)),
     :              rxpp,out2)
            endif
         endif

         vnew=0.d0
         wnew=0.d0
         ll=0
         if(cp+mbm.le.ftn0)then
            do i=cp+1,mcm-1
               ll=ll+1
               if (ip.eq.kp) then
                  call pppnergy ( potk, rxp(:,ll), ip,
     :                 rxp(:,ll), kp, i,
     :                 vvv, www )
               else
                  call pppnergy ( potk, rxp(:,ll), ip,
     :                 rxpp(:,ll), kp, i,
     :                 vvv, www )
               endif
               vnew=vnew+vvv
               wnew=wnew+www
            enddo
         else
            do i=cp+1,ftn0
               ll=ll+1
               if (ip.eq.kp) then
                  call pppnergy ( potk, rxp(:,ll), ip,
     :                 rxp(:,ll), kp, i,
     :                 vvv, www )
               else
                  call pppnergy ( potk, rxp(:,ll), ip,
     :                 rxpp(:,ll), kp, i,
     :                 vvv, www )
               endif
               vnew=vnew+vvv
               wnew=wnew+www
            enddo
            do i=1,mcm-1
               ll=ll+1
               if (ip.eq.kp) then
                  call pppnergy ( potk, rxp(:,ll), nip,
     :                 rxp(:,ll), nkp, i,
     :                 vvv, www )
               else
                  call pppnergy ( potk, rxpp(:,ll), nip,
     :                 rxp(:,ll), nkp, i,
     :                 vvv, www )
               endif
               vnew=vnew+vvv
               wnew=wnew+www
            enddo
         endif

         ps=ps+vnew-vold

         call acc_p(ps,ip,kp,cp)
         if (ps.gt.ranf(dummy)) then

            acatmab = acatmab + 1.d0
            call update(cp,rxp,ip,nkp)
            if (ip.ne.kp) then
               call update(cp,rxpp,kp,nip)
               call swap(cp,ip,nip,kp,nkp)
               call switch(next(ip),next(kp))
               call switch(mext(cp,ip),mext(cp,kp))
               acatmas = acatmas + 1.d0
            endif

c     restricted path integral
            do i=1,ftn0
               call ifdm(isp(ip),rs,i,ifdms)
               if ( ifdms .lt. 0. ) goto 1234
            enddo

            kenew=0.d0
            do k=1,np
               if(k.eq.ip.or.k.eq.kp.or.k.eq.nip.or.k.eq.nkp)then
                  do i=1,ftn0
                     call kkknergy ( rx(:,i,k), k, i,
     :                    kkk )
                     kenew=kenew+kkk
                  enddo
               endif
            enddo
            ke=ke+kenew-keold

            v=v+vnew-vold
            w=w+wnew-wold
         endif
         goto 1235
 1234    continue
         rx=rxold
         acatmab = acatmab - 1.0
         if (ip.ne.kp) then
            acatmas = acatmas - 1.0
            call switch(next(ip),next(kp))
            call switch(mext(cp,ip),mext(cp,kp))
         endif
 1235    continue

c     build the permutations cycles in lext

         call permcyc()

         IF (STEP.GT.IEQUI) THEN
            ACM = ACM + 1.0

C     ** CALCULATE INSTANTANEOUS VALUES **

               IF (POTK .EQ. 'LJ') THEN
```

```
                 VN = ( V + VLRC )
              ELSE
                 VN = V
                 WN = W
              ENDIF
              CALL CWIND( ISP(IP), WIND )

C     ** ACCUMULATE AVERAGES **

           ACV    = ACV    + VN
           ACVSQ  = ACVSQ  + VN*VN
           ACW    = ACW    + WN
           ACWSQ  = ACWSQ  + WN*WN
           ACKE   = ACKE   + KE
           ACKESQ = ACKESQ + KE*KE
           DO J=1,NMIX
              DO I=1,DIM
                 ACWINDSQ(J,I) = ACWINDSQ(J,I) + WIND(J,I)**2.
              ENDDO
           ENDDO
        ENDIF

        IF (IFBRIDGE) GOTO 97

C     ****************************************************************
C     ** ENDS BRIDGE&SWAP MOVE                                      **
C     ****************************************************************

 77     CONTINUE

C     ****************************************************************
C     ** DISPLACEMENT MOVE                                          **
C     ****************************************************************

C     ** PREVIOUS IP **

        DO I=1,NP
           J=NEXT(I)
           PREV(J)=I
        ENDDO

        NIP = NEXT(IP)
        PIP = PREV(IP)

        DO 90 C1 = 1, FTNO

           DO IDIM = 1, DIM
              RXIOLD(IDIM) = RX(IDIM,C1,IP)
           ENDDO

C     ** CALCULATE THE ENERGY OF I IN THE OLD CONFIGURATION **

           CALL PENERGY ( POTK, RXIOLD, IP, C1,
     :            VOLD, WOLD )

           CALL KENERGY ( RXIOLD, IP, C1,
     :            KEOLD )

C     ** INSTANTANEOUS VALUE OF THE ACTION **

           ACTOLD = KEOLD + VOLD

C     ** MOVE I AND PICKUP THE CENTRAL IMAGE **

           DO IDIM = 1, DIM
              RXINEW(IDIM) = RXIOLD(IDIM) +
     :             ( 2.0 * RANF ( DUMMY ) - 1.0 )*DRMAX
c              RXINEW(IDIM) = RXINEW(IDIM) -
c     :                DNINT ( RXINEW(IDIM)/SIGMA )*SIGMA
           ENDDO

C     ** CALCULATE THE ENERGY OF I IN THE NEW CONFIGURATION **

           CALL PENERGY ( POTK, RXINEW, IP, C1,
     :            VNEW, WNEW )

           CALL KENERGY ( RXINEW, IP, C1,
     :            KENEW )

C     ** INSTANTANEOUS VALUE OF THE ACTION **

           ACTNEW = KENEW + VNEW

           DELTV  = VNEW  - VOLD
           DELTW  = WNEW  - WOLD
           DELTKE = KENEW - KEOLD

C     ** CHECK FOR ACCEPTANCE **

           DELTACT  = ACTNEW - ACTOLD
           DELTACTB = LS*DELTACT

           IF ( EXP ( - DELTACTB ) .GT. RANF ( DUMMY ) ) THEN
              V          = V  + DELTV
              W          = W  + DELTW
              KE         = KE + DELTKE
              ACATMA = ACATMA + 1.0
              DO IDIM = 1, DIM
                 RX(IDIM,C1,IP)  = RXINEW(IDIM)
C     IMAGINARY TIME PERIODIC BOUNDARY CONDITIONS
                 IF (C1.EQ.1) THEN
                    RX(IDIM,FTNO+1,PIP)=RX(IDIM,C1,IP)
                 ENDIF
                 IF (C1.EQ.FTNO) THEN
                    RX(IDIM,0,NIP)=RX(IDIM,C1,IP)
                 ENDIF
              ENDDO

C     RESTRICTED PATH INTEGRAL
              DO I=1,FTNO
                 CALL IFDM(ISP(IP),RS,I,IFDMS)
                 IF ( IFDMS .LT. 0. ) GOTO 1236
              ENDDO
              GOTO 1237
 1236         CONTINUE
              V          = V  - DELTV
              W          = W  - DELTW
              KE         = KE - DELTKE
              ACATMA = ACATMA - 1.0

              DO IDIM = 1, DIM
                 RX(IDIM,C1,IP)  = RXIOLD(IDIM)
                 IF (C1.EQ.1) THEN
                    RX(IDIM,FTNO+1,PIP)=RX(IDIM,C1,IP)
                 ENDIF
                 IF (C1.EQ.FTNO) THEN
                    RX(IDIM,0,NIP)=RX(IDIM,C1,IP)
                 ENDIF
              ENDDO
 1237         CONTINUE
           ENDIF

           IF (STEP.GT.IEQUI) THEN
              ACM = ACM + 1.0

C     ** CALCULATE INSTANTANEOUS VALUES **

C     HISTOGRAM FOR g(r)
              IF ( MOD ( C1, ICALCG ) .EQ. 0 ) THEN
                 CALL GOFR(C1, DENS, MAXBIN, HIST11, HIST12, HIST22)
              ENDIF

              IF (POTK .EQ. 'LJ') THEN
                 VN = ( V + VLRC )
              ELSE
                 VN = V
                 WN = W
              ENDIF
              CALL CWIND( ISP(IP),WIND )

C     ** ACCUMULATE AVERAGES **

              ACV    = ACV    + VN
              ACVSQ  = ACVSQ  + VN*VN
              ACW    = ACW    + WN
              ACWSQ  = ACWSQ  + WN*WN
              ACKE   = ACKE   + KE
              ACKESQ = ACKESQ + KE*KE
              DO J=1,NMIX
                 DO I=1,DIM
                    ACWINDSQ(J,I) = ACWINDSQ(J,I) + WIND(J,I)**2.
                 ENDDO
              ENDDO
           ENDIF

C     ****************************************************************
C     ** END DISPLACEMENT MOVE                                      **
C     ****************************************************************

 90     CONTINUE

C     ****************************************************************
C     ** ENDS LOOP OVER TIME SLICES                                 **
C     ****************************************************************

 97     CONTINUE

C     ** WRITE OUT THE INSTANTANEOUS VALUES ON FORT.9 **

        SF  = 0.0
        SFI = 0.0
        DO I=1, NMIX
           DO IDIM = 1, DIM
              SFI(I) = SFI(I) + WIND(I,IDIM)*WIND(I,IDIM)
              SF(I)  = SF(I)  + ACWINDSQ(I,IDIM)
           ENDDO
        ENDDO
        DO I=1, NMIX
           SFI(I) = SFI(I)*FTM(I)/(DIM*BETA*NPM(I))
           SF(I)  = SF(I)*FTM(I)/(DIM*BETA*NPM(I)*ACM)
        ENDDO
        IF (ISP(IP).EQ.1)THEN
           WSP(1)=1
           WSP(2)=0
        ELSE
           WSP(1)=0
           WSP(2)=1
        ENDIF

        CALL FLUSH(9)
        IF ( MOD ( STEP, IPRINT ) .EQ. 0 ) THEN
           WRITE(9,*) STEP,DIM*NP/2/LS-KE/FTNO,V/FTNO,W/FTNO,
     :                SFI,WSP
C           WRITE(9,*) STEP, DIM*NP/2/LS-KE/FTNO, V/FTNO, W/FTNO
        ENDIF

C     ** CREATE POSITION DISTRIBUTION **

C          CALL DISTR(30_8,STEP*FTNO*NP/30)

C     ** PERFORM PERIODIC OPERATIONS **

        IF ( MOD ( STEP, IRATIO ) .EQ. 0 ) THEN

C     ** ADJUST MAXIMUM DISPLACEMENT **

           RATIO  = ACATMA  / DBLE ( FTNO * IRATIO )
           RATIOB = ACATMAB / DBLE ( IRATIO )
           RATIOS = ACATMAS / DBLE ( IRATIO )

           IF ( RATIO .GT. 0.5 ) THEN
```

```
C                DRMAX  = DRMAX  * 1.05

            ELSE

C                DRMAX  = DRMAX  * 0.95

            ENDIF

            ACATMA  = 0.0
            ACATMAB = 0.0
            ACATMAS = 0.0

         ENDIF

         IF ( MOD ( STEP, IPRINT ) .EQ. 0 ) THEN

C     ** WRITE OUT RUNTIME INFORMATION **

            WRITE(*,*) '          step            acm  dr/sig    ard
     :     arb       ars'
            WRITE(*,'(2(2XI13),(2Xf6.4),4(2Xf8.6))') STEP, INT(ACM),
     :                                 ALPHA, RATIO, RATIOB, RATIOS
            WRITE(*,*) '            ke              v              w
     :        sf'
            WRITE(*,'(4(2XE13.7))') KE/FTN0, V/FTN0, W/FTN0, SFI
            WRITE(*,*) '          <ke>            <v>            <w>
     :         <sf>'
            WRITE(*,'(4(2XE13.7))') ACKE/ACM/FTN0, ACV/ACM/FTN0,
     :                              ACW/ACM/FTN0, SF

         ENDIF

         IF ( MOD ( STEP, ISAVE ) .EQ. 0 ) THEN

C     ** WRITE OUT THE CONFIGURATION AT INTERVALS **

            CALL WRITCN ( CNFILE )
            REWIND(20)
            CALL FLUSH(20)
            WRITE(20,*)(HIST11(I),I=1,MAXBIN),
     &                 (HIST12(I),I=1,MAXBIN),
     &                 (HIST22(I),I=1,MAXBIN),
     &           ACV,ACVSQ,ACW,ACWSQ,ACKE,ACKESQ,ACWINDSQ,ACM,STEP

C     ** WRITE OUT THE g(r) **
            IF ( STEP .GT. IEQUI ) THEN
               TSTEP = DBLE((STEP - IEQUI)*INT(FTN0/ICALCG))
               REWIND(11)
               REWIND(12)
               REWIND(13)
               REWIND(14)
               DO BIN = 1, MAXBIN2
                  RLOWER = DBLE ( BIN - 1 ) * DELR
                  RUPPER = RLOWER + DELR
                  NIDEAL11 = CONST11*( RUPPER**DIM - RLOWER**DIM )
                  NIDEAL12 = CONST12*( RUPPER**DIM - RLOWER**DIM )
                  NIDEAL22 = CONST22*( RUPPER**DIM - RLOWER**DIM )
                  GR11 = DBLE(HIST11(BIN))/TSTEP/DBLE(NPM(1))/NIDEAL11
                  GR12 = DBLE(HIST12(BIN))/TSTEP/DBLE(NPM(1))/NIDEAL12
                  GR22 = DBLE(HIST22(BIN))/TSTEP/DBLE(NPM(2))/NIDEAL22
                  CALL FLUSH(11)
                  CALL FLUSH(12)
                  CALL FLUSH(13)
                  CALL FLUSH(14)
                  WRITE (11,*) DBLE(BIN-1)*DELR, GR11
                  WRITE (12,*) DBLE(BIN-1)*DELR, GR12
                  WRITE (13,*) DBLE(BIN-1)*DELR, GR22
                  WRITE (14,*) DBLE(BIN-1)*DELR,
     :            CO(1)**2*GR11+2.d0*CO(1)*CO(2)*GR12+CO(2)**2*GR22
               ENDDO
            ENDIF

         ENDIF

 100     CONTINUE

C     ****************************************************************
C     ** ENDS THE LOOP OVER CYCLES                                  **
C     ****************************************************************

         WRITE(*,'(//'' END OF MARKOV CHAIN          ''//)')

C     ** CHECKS FINAL VALUE OF THE POTENTIAL ENERGY IS CONSISTENT **

         CALL SUMUP (POTK, OVRLAP, KEEND, VEND, WEND)

         IF ( ABS(VEND - V) .GT. 1.0d-03 ) THEN

            WRITE(*,'(''PROBLEM WITH V ENERGY !!!'')')
            WRITE(*,'('' VEND  = '', E20.6)') VEND
            WRITE(*,'('' V     = '', E20.6)') V

         ENDIF

         IF ( ABS(KEEND - KE) .GT. 1.0d-03 ) THEN

            WRITE(*,'(''PROBLEM WITH KE ENERGY !!!'')')
            WRITE(*,'('' KEEND  = '', E20.6)') KEEND
            WRITE(*,'('' KE     = '', E20.6)') KE

         ENDIF

C     ** WRITE OUT THE FINAL CONFIGURATION FROM THE RUN **

         CALL WRITCN ( CNFILE )

C     ** CALCULATE AND WRITE OUT RUNNING AVERAGES **

         AVV   = ACV / ACM
         ACVSQ = ( ACVSQ / ACM ) - AVV ** 2
         AVKE   = ACKE / ACM
         ACKESQ = ( ACKESQ / ACM ) - AVKE ** 2

C    ** CALCULATE FLUCTUATIONS **

         IF ( ACVSQ  .GT. 0.0 ) FLV  = SQRT ( ACVSQ/ACM )/FTN0
         IF ( ACKESQ .GT. 0.0 ) FLKE = SQRT ( ACKESQ/ACM )/FTN0

         WRITE(*,'(/'' AVERAGES ''/ )')
         WRITE(*,'('' <V/N>                 = '',E12.6)') AVV/FTN0
         WRITE(*,'('' <KE/N>                = '',E12.6)') AVKE/FTN0

         WRITE(*,'(/'' FLUCTUATIONS ''/)')

         WRITE(*,'('' FLUCTUATION IN <V/N>  = '',E12.6)') FLV
         WRITE(*,'('' FLUCTUATION IN <KE/N> = '',E12.6)') FLKE
         WRITE(*,'(/'' END OF SIMULATION ''/)')

         CLOSE (UNIT = 9 )
         CLOSE (UNIT = 11)
         CLOSE (UNIT = 12)
         CLOSE (UNIT = 13)
         CLOSE (UNIT = 14)
C     ****************************************************************
C     ****************************************************************
C     ****************************************************************
C     ** THE END                                                    **
C     ****************************************************************
C     ****************************************************************
C     ****************************************************************
         STOP
         END


         subroutine permcyc()
         implicit none
!     given a particle i lext(i) returns the cycle it belongs to
         INCLUDE       'mc-mix.par'

         integer*8 i,j,k,ii(mnp),jj,kk,ll

         kk=1
         do i=1,np
            ll=1
            ii(ll)=i
 1          jj=next(ii(ll))
            do j=1,ll
               if(jj.eq.ii(j))goto 2
            enddo
            ll=ll+1
            ii(ll)=jj
            goto 1
 2          do k=1,ll
               lext(ii(k))=kk
            enddo
            kk=kk+1
         enddo

         return
         end


         subroutine switch(i,j)
!     switch i and j
         implicit none
         integer*8 i,j,k
         k=i
         i=j
         j=k
         return
         end


         subroutine acc_p(p,ip,kp,j)
         implicit none
!     complete acceptance probability
         INCLUDE       'mc-mix.par'

         integer*8 ip,kp,j,idim
         real*8 p, rho
         real*8 rxink,rxnik
         real*8 rxini,rxknk
         integer*8 mcm

         p=exp(-ls*p)         ! contribution from the pair potential
         if (ip.eq.kp) return

         mcm = j+mbm-floor((j+mbm-.1)/ftn0)*ftn0
         rho=0.d0
         do idim=1,dim
            if(j+mbm.le.ftn0)then
               rxink=rx(idim,j,ip)-rx(idim,mcm,kp)
               rxnik=rx(idim,j,kp)-rx(idim,mcm,ip)
               rxini=rx(idim,j,ip)-rx(idim,mcm,ip)
               rxknk=rx(idim,j,kp)-rx(idim,mcm,kp)
               rxink=rxink-dnint(rxink/sigma)*sigma
               rxnik=rxnik-dnint(rxnik/sigma)*sigma
               rxini=rxini-dnint(rxini/sigma)*sigma
               rxknk=rxknk-dnint(rxknk/sigma)*sigma
            else
               rxink=rx(idim,j,ip)-rx(idim,mcm,next(kp))
               rxnik=rx(idim,j,kp)-rx(idim,mcm,next(ip))
               rxini=rx(idim,j,ip)-rx(idim,mcm,next(ip))
               rxknk=rx(idim,j,kp)-rx(idim,mcm,next(kp))
               rxink=rxink-dnint(rxink/sigma)*sigma
               rxnik=rxnik-dnint(rxnik/sigma)*sigma
               rxini=rxini-dnint(rxini/sigma)*sigma
               rxknk=rxknk-dnint(rxknk/sigma)*sigma
            endif
            rho=rho+rxink**2+rxnik**2-rxini**2-rxknk**2
         enddo
         rho=ftm(isp(ip))*rho/(2.*mbm*ls)
```

```
      p=p*exp(-rho)
      return
      end


      subroutine update(j,rxp,ip,nip)
      implicit none
!     updates a portion of the current path x using the proposed path xp
      INCLUDE       'mc-mix.par'

      integer*8 ip,nip,kp,nkp,j,l,k,idim
      integer*8 mcm
      real*8 rxp(mdim,0:n)

      mcm = j+mbm-floor((j+mbm-.1)/ftn0)*ftn0

      l=0
      if(j+mbm.le.ftn0)then
         do k=j+1,mcm-1
            l=l+1
            do idim=1,dim
               rx(idim,k,ip)=rxp(idim,l)
            enddo
         enddo
      else
         do k=j+1,ftn0
            l=l+1
            do idim=1,dim
               rx(idim,k,ip)=rxp(idim,l)
            enddo
         enddo
         do k=1,mcm-1
            l=l+1
            do idim=1,dim
               rx(idim,k,nip)=rxp(idim,l)
            enddo
         enddo
      endif
      do idim=1,dim
         rx(idim,ftn0+1,ip)=rx(idim,1,nip)
         rx(idim,0,nip)=rx(idim,ftn0,ip)
      enddo

      return
      end


      subroutine swap(j,ip,nip,kp,nkp)
      implicit none
!     updates a portion of the current path x using the proposed path xp
      INCLUDE       'mc-mix.par'

      integer*8 ip,nip,kp,nkp,j,k,idim
      integer*8 mcm
      real*8    rr

      mcm = j+mbm-floor((j+mbm-.1)/ftn0)*ftn0

      if(j+mbm.le.ftn0)then
         do k=mcm,ftn0
            do idim=1,dim
               rr=rx(idim,k,ip)
               rx(idim,k,ip)=rx(idim,k,kp)
               rx(idim,k,kp)=rr
            enddo
         enddo
         do idim=1,dim
            rx(idim,ftn0+1,ip)=rx(idim,1,nkp)
            rx(idim,ftn0+1,kp)=rx(idim,1,nip)
            rx(idim,0,nip)=rx(idim,ftn0,kp)
            rx(idim,0,nkp)=rx(idim,ftn0,ip)
         enddo
      endif
      return
      end


      subroutine bridge(x0,x1,std,xnew,out)
      implicit none
!     sample m gaussians with std from xnew(0)=x0 to xnew(ftn0)=x1
      INCLUDE       'mc-mix.par'

      integer*8 l1,l2,l3,j,out,idim
      real*8 std,d,s,xi
      real*8 x0(mdim),x1(mdim),xnew(mdim,0:n)

      out=0
      l3=mbm
      do idim=1,dim
         xnew(idim,0)=x0(idim)
         xnew(idim,l3)=x0(idim)+((x1(idim)-x0(idim))-
     :          dnint((x1(idim)-x0(idim))/sigma)*sigma)
      enddo
      do j=1,mbm-1
         l1=j-1
         l2=j
         s=std*(dble(l3-l2)/dble(l3-l1))**0.5d0
         do idim=1,dim
            d=xnew(idim,l3)-xnew(idim,l1)
            d=d-dnint(d/sigma)*sigma
            xnew(idim,j)=xnew(idim,l1)+d/dble(l3-l1)+xi(s)
c            if (xnew(idim,j).gt.sigma/2.or.
c     :             xnew(idim,j).lt.-sigma/2) then
c               out=1
c               return
c            endif
c            xnew(idim,j)=xnew(idim,j)-dnint(xnew(idim,j)/sigma)*sigma
         enddo
      enddo
      return
      end


      function xi(std)
!     sample a gaussian with standard deviation std (box-muller method)
      implicit none
      real*8 xi,std,pi,ranf
      data pi/3.14159265358979323846264338328d0/
      xi=cos(pi*ranf(0.d0))*std*sqrt(-2.d0*log(tiny(pi)+ranf(0.d0)))
      return
      end


      SUBROUTINE DISTR(NN,NORM)
      IMPLICIT NONE
C     WRITES ON FORT.10 THE X-POSITION DISTRIBUTION
      INCLUDE       'mc-mix.par'

      REAL*8        DS,DIST(0:1000)
      INTEGER*8     NN,NORM,I,J,K
      SAVE          DIST

      DS=SIGMA/NN

      DO I=0,NN
         DO J=1,FTN0
            DO K=1,NP
               IF(-SIGMA/2+(I-.5)*DS.LT.RX(1,J,K).AND.
     :              RX(1,J,K).LT.-SIGMA/2+(I+.5)*DS) THEN
                 DIST(I)=DIST(I)+1.D0
               ENDIF
            ENDDO
         ENDDO
         WRITE(10,*) -SIGMA/2+I*DS,DIST(I)/NORM
      ENDDO

      CLOSE(UNIT=10)

      RETURN
      END


      FUNCTION FACT ( N )
      IMPLICIT NONE
C     FACTORIAL FUNCTION
      INTEGER*8 FACT,N,P,I
      P=1
      DO I=1,N
         P=P*I
      ENDDO
      FACT=P
      END


      SUBROUTINE SUMUP (POTK, OVRLAP, KE, V, W)
      IMPLICIT NONE
C     ******************************************************************
C     ** CALCULATES THE TOTAL ENERGY                                  **
C     **                                                              **
C     ** USAGE:                                                       **
C     **                                                              **
C     ** THE SUBROUTINE RETURNS THE TOTAL ENERGY AT THE               **
C     ** BEGINNING AND END OF THE RUN.                                **
C     ******************************************************************
      INCLUDE       'mc-mix.par'

      REAL*8        V, KE, VV, KK
      LOGICAL       OVRLAP
      CHARACTER     POTK*(*)

      REAL*8        RXII, RXIJ
      REAL*8        VIJ, WIJ, RIJSQ, W, WW

      INTEGER*8     TAU, I, J, IDIM

C     POTENTIAL ACTION

      VV      = 0.0
      WW      = 0.0

C     ** LOOP OVER ALL THE PAIRS IN THE LIQUID **

      DO TAU = 1, FTN0
         DO 100 I = 1, NP - 1
            DO 99 J = I + 1, NP
               RIJSQ = 0.d0
               DO IDIM = 1, DIM
                  RXIJ = RX(IDIM,TAU,I) - RX(IDIM,TAU,J)
C     ** MINIMUM IMAGE THE PAIR SEPARATIONS **
                  RXIJ = RXIJ -
     :                 DNINT ( RXIJ/SIGMA )*SIGMA
                  RIJSQ = RIJSQ + RXIJ * RXIJ
               ENDDO

               CALL POT (I, J, RIJSQ, VIJ, WIJ, POTK)
               VV   = VV + VIJ
               WW   = WW + WIJ
 99         CONTINUE
 100     CONTINUE
         V=VV
         W=WW
      ENDDO

C     KINETIC ACTION

      KK     = 0.0

      DO TAU = 1, FTN0
         DO I = 1, NP
            DO IDIM = 1, DIM
               RXII = RX(IDIM,TAU,I)-RX(IDIM,TAU+1,I)
               RXII = RXII -
```

```
     :               DNINT ( RXII/SIGMA )*SIGMA
                KK=KK+FTM(ISP(I))*(RXII**2.)/(2.DO*LS**2.)
             ENDDO
          ENDDO
          KE=KK
       ENDDO

       RETURN
       END


       SUBROUTINE PENERGY ( POTK, RXI, I, TAU,
     :    V, W )
       IMPLICIT NONE
C    ****************************************************************
C    ** RETURNS THE POTENTIAL ENERGY OF ATOM I WITH ALL OTHER ATOMS.  **
C    **                                                              **
C    ** PRINCIPAL VARIABLES:                                         **
C    **                                                              **
C    ** INTEGER I                 THE ATOM OF INTEREST               **
C    ** INTEGER NP                THE NUMBER OF ATOMS                **
C    ** INTEGER TAU               THE TIMESTEP                       **
C    ** REAL*8    RX, RY, RZ        THE ATOM POSITIONS               **
C    ** REAL*8    RXI,RYI,RZI       THE COORDINATES OF ATOM I        **
C    ** REAL*8    V                 THE POTENTIAL ENERGY OF ATOM I   **
C    ** REAL*8    W                 THE VIRIAL OF ATOM I             **
C    **                                                              **
C    ** USAGE:                                                       **
C    **                                                              **
C    ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY     **
C    ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND        **
C    ** AFTER THE RANDOM DISPLACEMENT OF I.                          **
C    ****************************************************************
       INCLUDE     'mc-mix.par'

       REAL*8        RXI(MDIM), V, W
       INTEGER*8     I, J, TAU, IDIM
       CHARACTER     POTK*(*)

       REAL*8        RXIJ, RIJSQ, VIJ, WIJ

       V      = 0.0
       W      = 0.0

C    ** LOOP OVER ALL MOLECULES EXCEPT I  **

       DO 100 J = 1, NP

          IF ( I .NE. J ) THEN
             RIJSQ = 0.d0
             DO IDIM = 1, DIM
                RXIJ = RXI(IDIM) - RX(IDIM,TAU,J)

                RXIJ = RXIJ -
     :               DNINT ( RXIJ/SIGMA )*SIGMA
                RIJSQ = RIJSQ + RXIJ * RXIJ
             ENDDO

             CALL POT (I, J, RIJSQ, VIJ, WIJ, POTK)
             V   = V + VIJ
             W   = W + WIJ

 90       ENDIF

100    CONTINUE

       RETURN
       END


       SUBROUTINE PPPNERGY ( POTK, RXI, I, RXJ, J,
     :    TAU, V, W )
       IMPLICIT NONE
C    ****************************************************************
C    ** RETURNS THE POTENTIAL ENERGY OF ATOMS I AND J WITH            **
C    ** ALL OTHER ATOMS PLUS THE ONE BETWEEN I AND J                  **
C    **                                                              **
C    ** PRINCIPAL VARIABLES:                                         **
C    **                                                              **
C    ** INTEGER I, J              THE ATOMS OF INTEREST              **
C    ** INTEGER NP                THE NUMBER OF ATOMS                **
C    ** INTEGER TAU               THE TIMESTEP                       **
C    ** REAL*8    RX, RY, RZ        THE ATOM POSITIONS               **
C    ** REAL*8    RXI,RYI,RZI       THE COORDINATES OF ATOM I        **
C    ** REAL*8    RXJ,RYJ,RZJ       THE COORDINATES OF ATOM J        **
C    ** REAL*8    V                 THE POTENTIAL ENERGY OF ATOM I   **
C    ** REAL*8    W                 THE VIRIAL OF ATOM I             **
C    **                                                              **
C    ** USAGE:                                                       **
C    **                                                              **
C    ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY     **
C    ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND        **
C    ** AFTER THE RANDOM DISPLACEMENT OF I.                          **
C    ****************************************************************
       INCLUDE     'mc-mix.par'

       REAL*8        RXI(MDIM), RXJ(MDIM), V, W
       INTEGER*8     I, J, K, TAU, IDIM
       CHARACTER     POTK*(*)

       REAL*8        RXIJ, RIJSQ, VIJ, WIJ

       V      = 0.0
       W      = 0.0

C    ** LOOP OVER ALL MOLECULES EXCEPT I AND J **

       DO 100 K = 1, NP
          IF ( K .NE. I .AND. K .NE. J ) THEN
             RIJSQ = 0.d0
             DO IDIM = 1, DIM
                RXIJ = RXI(IDIM) - RX(IDIM,TAU,K)

                RXIJ = RXIJ -
     :               DNINT ( RXIJ/SIGMA )*SIGMA
                RIJSQ = RIJSQ + RXIJ * RXIJ
             ENDDO

             CALL POT (I, K, RIJSQ, VIJ, WIJ, POTK)
             V   = V + VIJ
             W   = W + WIJ

             IF ( I .NE. J ) THEN
                RIJSQ = 0.d0
                DO IDIM = 1, DIM
                   RXIJ = RXJ(IDIM) - RX(IDIM,TAU,K)

                   RXIJ = RXIJ -
     :                 DNINT ( RXIJ/SIGMA )*SIGMA
                   RIJSQ = RIJSQ + RXIJ * RXIJ
                ENDDO

                CALL POT (K, J, RIJSQ, VIJ, WIJ, POTK)
                V   = V + VIJ
                W   = W + WIJ

             ENDIF

          ENDIF

 100    CONTINUE

C        RETURN

       IF (I .NE. J) THEN
          RIJSQ = 0.d0
          DO IDIM = 1, DIM
             RXIJ = RXI(IDIM) - RXJ(IDIM)

             RXIJ = RXIJ -
     :              DNINT ( RXIJ/SIGMA )*SIGMA
             RIJSQ = RIJSQ + RXIJ * RXIJ
          ENDDO

          CALL POT (I, J, RIJSQ, VIJ, WIJ, POTK)
          V   = V + VIJ
          W   = W + WIJ
       ENDIF

       RETURN
       END


       SUBROUTINE KENERGY ( RXI, I, TAU, KE )
       IMPLICIT NONE
C    ****************************************************************
C    ** RETURNS THE KINETIC ENERGY OF ATOM I WITH ALL OTHER ATOMS.    **
C    **                                                              **
C    ** PRINCIPAL VARIABLES:                                         **
C    **                                                              **
C    ** INTEGER I                 THE ATOM OF INTEREST               **
C    ** INTEGER NP                THE NUMBER OF ATOMS                **
C    ** INTEGER TAU               THE TIMESTEP                       **
C    ** REAL*8    RX, RY, RZ        THE ATOM POSITIONS               **
C    ** REAL*8    RXI,RYI,RZI       THE COORDINATES OF ATOM I        **
C    ** REAL*8    KE                THE KINETIC ENERGY OF ATOM I     **
C    **                                                              **
C    ** USAGE:                                                       **
C    **                                                              **
C    ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY     **
C    ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND        **
C    ** AFTER THE RANDOM DISPLACEMENT OF I.                          **
C    ****************************************************************
       INCLUDE     'mc-mix.par'

       REAL*8        RXI(MDIM), KE
       REAL*8        RXIP, RXIS, RXII
       INTEGER*8     I, TAU, IDIM

       KE = 0.d0
       DO IDIM = 1, DIM
          RXIP = RX(IDIM,TAU-1,I)
          RXIS = RX(IDIM,TAU+1,I)
          RXII = RXI(IDIM) - RXIP
          RXII = RXII - DNINT ( RXII/SIGMA )*SIGMA
          KE=KE+0.5*FTM(ISP(I))*(RXII/LS)**2.
          RXII = RXI(IDIM) - RXIS
          RXII = RXII - DNINT ( RXII/SIGMA )*SIGMA
          KE=KE+0.5*FTM(ISP(I))*(RXII/LS)**2.
       ENDDO

       RETURN
       END


       SUBROUTINE KKKNERGY ( RXI, I, TAU, KE )
       IMPLICIT NONE
C    ****************************************************************
C    ** RETURNS THE KINETIC ENERGY OF ATOM I WITH ALL OTHER ATOMS.    **
C    **                                                              **
C    ** PRINCIPAL VARIABLES:                                         **
C    **                                                              **
C    ** INTEGER I                 THE ATOM OF INTEREST               **
C    ** INTEGER NP                THE NUMBER OF ATOMS                **
C    ** INTEGER TAU               THE TIMESTEP                       **
C    ** REAL*8    RX, RY, RZ        THE ATOM POSITIONS               **
C    ** REAL*8    RXI,RYI,RZI       THE COORDINATES OF ATOM I        **
C    ** REAL*8    KE                THE KINETIC ENERGY OF ATOM I     **
C    **                                                              **
C    ** USAGE:                                                       **
C    **                                                              **
C    ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY     **
```

```
C    ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND        **
C    ** AFTER THE RANDOM DISPLACEMENT OF I.                            **
C    ***********************************************************************
      INCLUDE     'mc-mix.par'

      REAL*8        RXI(MDIM), KE
      REAL*8        RXIS, RXII
      INTEGER*8     I, TAU, IDIM

      KE = 0.d0
      DO IDIM = 1, DIM
         RXIS = RX(IDIM,TAU+1,I)
         RXII = RXI(IDIM) - RXIS
         RXII = RXII - DNINT ( RXII/SIGMA )*SIGMA
         KE=KE+0.5*FTM(ISP(I))*(RXII/LS)**2.
      ENDDO

      RETURN
      END


      SUBROUTINE CWIND ( ISPEC, WIND )
      IMPLICIT NONE
C    ***********************************************************************
C    ** RETURNS THE WINDING NUMBER.                                     **
C    ***********************************************************************
      INCLUDE     'mc-mix.par'

      REAL*8        WIND(MMIX,MDIM), WI, RX1, RX2
      INTEGER*8     I, J, IDIM, ISPEC

      WIND = 0.d0
      DO I = 1, NP
         IF (ISP(I).NE.ISPEC) GOTO 10
         DO IDIM = 1, DIM
            DO J = 1, FTN0 - 1
               RX1 = RX(IDIM,J  ,I)
               RX2 = RX(IDIM,J+1,I)
               WI  = RX2-RX1-DNINT((RX2-RX1)/SIGMA)*SIGMA
               WIND(ISPEC,IDIM) = WIND(ISPEC,IDIM) + WI
            ENDDO
         ENDDO
 10      CONTINUE
      ENDDO

      RETURN
      END


      REAL*8 FUNCTION RANF ( DUMMY )
      IMPLICIT NONE
C    ***********************************************************************
C    ** RETURNS A UNIFORM RANDOM VARIATE IN THE RANGE 0 TO 1.           **
C    **                                                                 **
C    **                 ***************                                 **
C    **                 **  WARNING  **                                 **
C    **                 ***************                                 **
C    **                                                                 **
C    ** GOOD RANDOM NUMBER GENERATORS ARE MACHINE SPECIFIC.             **
C    ** PLEASE USE THE ONE RECOMMENDED FOR YOUR MACHINE.                **
C    **                                                                 **
C    ** RAND(FLAG) returns a pseudo-random number from a uniform        **
C    ** distribution between 0 and 1. If FLAG is 0, the next number     **
C    ** in the current sequence is returned; if FLAG is 1, the          **
C    ** generator is restarted by CALL SRAND(0); if FLAG has any        **
C    ** other value, it is used as a new seed with SRAND.               **
C    ***********************************************************************
      INTEGER*8   L, C, M
      PARAMETER ( L = 1029, C = 221591, M = 1048576 )
      INTEGER*8   SEED
      SAVE        SEED
      DATA        SEED / 0 /
      REAL*8      DUMMY

C       SEED = MOD ( SEED * L + C, M )
C       RANF = DBLE( SEED ) / M

      RANF = RAND ( )

      RETURN
      END


      SUBROUTINE READCN ( CNFILE )
      IMPLICIT NONE
C    ***********************************************************************
C    ** SUBROUTINE TO READ IN THE CONFIGURATION FROM UNIT 10            **
C    ***********************************************************************
      INCLUDE     'mc-mix.par'

      CHARACTER   CNFILE*(*), HASH*1, MYFMT*77

      INTEGER*8   CNUNIT, I, J, NNP, NI, IDIM
      PARAMETER ( CNUNIT = 10 )

C   ***********************************************************************

      OPEN ( UNIT = CNUNIT, FILE = CNFILE, STATUS = 'OLD' )
      WRITE(MYFMT,'(A,I10,A)') '(I3,3X,',DIM,'(F12.6,3X),I3)'

      READ ( CNUNIT,* ) HASH, FTN0, NNP
      IF ( NNP .NE. NP ) STOP 'N ERROR IN READCN'

      DO 100 I = 1, NNP
         READ ( CNUNIT,* ) HASH, NEXT(I)
         NI = NEXT(I)
         DO 90 J = 1, FTN0+1
            READ ( CNUNIT, MYFMT) LEXT(I), (RX(IDIM,J,I),IDIM=1,DIM),
     :                            ISP(I)
 90      ENDDO
         READ ( CNUNIT,13 )
         READ ( CNUNIT,13 )
         DO IDIM = 1, DIM
            RX(IDIM,0,NI) = RX(IDIM,FTN0,I)
         ENDDO
 100   ENDDO
 13    FORMAT(A2)

      CLOSE ( UNIT = CNUNIT )

      RETURN
      END


      SUBROUTINE WRITCN ( CNFILE )
      IMPLICIT NONE
C    ***********************************************************************
C    ** SUBROUTINE TO WRITE OUT THE CONFIGURATION TO UNIT 10            **
C    ***********************************************************************
      INCLUDE     'mc-mix.par'

      CHARACTER   CNFILE*(*), MYFMT*77

      INTEGER*8   CNUNIT, I, J, IDIM
      PARAMETER ( CNUNIT = 10 )

C   ***********************************************************************

      OPEN ( UNIT = CNUNIT, FILE = CNFILE, STATUS = 'UNKNOWN' )
      WRITE(MYFMT,'(A,I10,A)') '(I3,3X,',DIM,'(F12.6,3X),I3)'

      WRITE(*,*) 'output to file -----------------------------------
     :---------------'
      CALL FLUSH ( CNUNIT )
      WRITE ( CNUNIT,* ) '#',FTN0,NP

      DO I = 1, NP
         WRITE ( CNUNIT,* ) '#', NEXT(I)
         DO J = 1, FTN0+1
            WRITE ( CNUNIT, MYFMT ) LEXT(I),
     :       (RX(IDIM,J,I)-DNINT(RX(IDIM,J,I)/SIGMA)*SIGMA,IDIM=1,DIM),
     :       ISP(I)
C              WRITE ( CNUNIT, MYFMT ) LEXT(I),
C     :        (RX(IDIM,J,I),IDIM=1,DIM), ISP(I)
         ENDDO
         WRITE ( CNUNIT,13 ) ''
         WRITE ( CNUNIT,13 ) ''
      ENDDO
 13   FORMAT(A2)

      CLOSE ( UNIT = CNUNIT )

      RETURN
      END


      SUBROUTINE INITCN ( CNFILE )
      IMPLICIT NONE
C    ***********************************************************************
C    ** SUBROUTINE TO INITIALIZE THE CONFIGURATION TO UNIT 10           **
C    ***********************************************************************
      INCLUDE     'mc-mix.par'

      REAL*8      RANF
      CHARACTER   CNFILE*(*), MYFMT*77

      INTEGER*8   CNUNIT, I, J, IDIM, I1, IP, NNP
      REAL*8      RXI(DIM), RXII(DIM, NP), RXIJ, RIJSQ
      PARAMETER ( CNUNIT = 10 )

C   ***********************************************************************

      OPEN ( UNIT = CNUNIT, FILE = CNFILE, STATUS = 'UNKNOWN' )
      WRITE(MYFMT,'(A,I10,A)') '(I3,3X,',DIM,'(F12.6,3X),I3)'

      WRITE ( CNUNIT,* ) '#',FTN0,NP
      RXII = 0.d0

      IP = 1
      NNP = NPM(IP)
      DO I = 1, NP
         ISP(I)=IP
         IF (I.GE.NNP) THEN
            IP = IP + 1
            NNP = NNP + NPM(IP)
         ENDIF
      ENDDO

      DO I = 1, NP
         WRITE ( CNUNIT,* ) '#', NEXT(I)
 10      DO IDIM = 1, DIM
            RXI(IDIM) = SIGMA*(RANF(0.d0)-.5)
         ENDDO
         DO I1 = 1, NP
            RIJSQ = 0.d0
            DO IDIM = 1, DIM
               RXIJ = RXI(IDIM)-RXII(IDIM,I1)
               RXIJ = RXIJ  -
     :              DNINT ( RXIJ/SIGMA )*SIGMA
               RIJSQ = RIJSQ + RXIJ * RXIJ
            ENDDO
            IF (RIJSQ.LE.EPSA(ISP(I),ISP(I1))**2) THEN
               GOTO 10
            ENDIF
         ENDDO
         DO IDIM = 1, DIM
            RXII(IDIM,I) = RXI(IDIM)
         ENDDO
         DO J = 1, FTN0+1
            WRITE ( CNUNIT, MYFMT ) I, (RXI(IDIM),IDIM=1,DIM), ISP(I)
            DO IDIM = 1, DIM
               RX(IDIM,J,I) = RXI(IDIM)
```

```
            ENDDO
         ENDDO
         WRITE ( CNUNIT,13 ) ''
         WRITE ( CNUNIT,13 ) ''
      ENDDO
 13   FORMAT(A2)

      CLOSE ( UNIT = CNUNIT )

      RETURN
      END


      SUBROUTINE POT (I, J, RIJSQ, VIJ, WIJ, POTK)
      IMPLICIT NONE
C    ****************************************************************
C    ** SUBROUTINE FOR THE PAIR POTENTIAL                          **
C    ****************************************************************
      INCLUDE     'mc-mix.par'

      INTEGER*8 I, J, K
      CHARACTER POTK*(*)
      REAL*8    RIJ, RIJSQ, RMIN, RMSQ, RCSQ, SR2, SR6, VIJ, WIJ
      REAL*8    COU3, QHS3, F
      PARAMETER ( COU3 = 4.760154727959107013287634700 57d0 )
      PARAMETER ( QHS3 = 24.d0 )

C       PI = ACOS(-1.d0)

      VIJ = 0.d0
      WIJ = 0.d0
      IF (DIM .GT. 3) THEN
         WRITE(*,*) "POTENITAIL CANNOT HAVE DIM > 3 !!!"
         STOP
      ENDIF
      IF (POTK .EQ. 'FREE') RETURN

      IF (POTK .EQ. 'LJ') THEN
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC
C     W. L. McMillan, Phys. Rev. A 138, 442 (1965)     C
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC
C          RMIN = TINY(PI)
         RMIN = 0.d0
         RIJSQ = RIJSQ / (SIG(I,J) * SIG(I,J))
         RMSQ = RMIN*RMIN
         RCSQ = RCUT*RCUT
         IF (RIJSQ .LT. RMSQ) THEN
            RIJSQ = RMSQ
            SR2 = 1.d0 / RIJSQ
            SR6 = SR2 * SR2 * SR2
            VIJ = SR6 * ( SR6 - 1.d0 )
            WIJ = SR6 * ( SR6 - 5.d-1 )
            VIJ = 4.d0 * EPS * VIJ
            WIJ = 48.d0 * EPS * WIJ / 3.d0
         ELSEIF (RIJSQ .GT. RMSQ .AND. RIJSQ .LT. RCSQ) THEN
            SR2 = 1.d0 / RIJSQ
            SR6 = SR2 * SR2 * SR2
            VIJ = SR6 * ( SR6 - 1.d0 )
            WIJ = SR6 * ( SR6 - 5.d-1 )
            VIJ = 4.d0 * EPS * VIJ
            WIJ = 48.d0 * EPS * WIJ / 3.d0
         ENDIF
      ELSEIF (POTK .EQ. 'BUMP') THEN
         RIJ = SQRT( RIJSQ )
         IF (RIJ .LE. SIG(I,J)) THEN
            VIJ = EPSR
         ENDIF
      ELSEIF (POTK .EQ. 'PSW') THEN
         RIJ = SQRT( RIJSQ )
         IF (RIJ .LE. SIG(I,J)) THEN
            VIJ = EPSR
         ELSEIF (RIJ .LE. SIG(I,J)+RCUT .AND. RIJ .GT. SIG(I,J)) THEN
            VIJ = -EPSA(I,J)
         ENDIF
      ELSEIF (POTK .EQ. 'COULOMBHS') THEN
c          F=0.
c          DO K=1,NMIX
c             F = F+SIG(1,K)*NPM(K)/(NPM(K)-1)
c          ENDDO
         RIJ = SQRT( RIJSQ )
         IF (RIJ .GT. EPSA(ISP(I),ISP(J))) THEN
            IF (DIM .EQ. 3) THEN
c               VIJ = -F*COU3*RCUT/SIGMA/2.
c               VIJ = VIJ + RCUT*/RIJ
               VIJ = SIG(ISP(I),ISP(J))*RCUT/RIJ
            ELSEIF (DIM .EQ. 2) THEN
c               VIJ = -F*(6.-PI+LOG(4.)-4.*LOG(SIGMA/RCUT))/4.
c               VIJ = VIJ - LOG(RIJ/RCUT)
               VIJ = -SIG(ISP(I),ISP(J))*LOG(RIJ/RCUT)
            ELSEIF (DIM .EQ. 1) THEN
c               VIJ = F*SIGMA/RCUT/4.
c               VIJ = VIJ - RIJ/RCUT
               VIJ = -SIG(ISP(I),ISP(J))*RIJ/RCUT
            ENDIF
         ELSE
            VIJ = 10.**10.
         ENDIF
      ELSEIF (POTK .EQ. 'COULOMB') THEN
c          F=0.
c          DO K=1,NMIX
c             F = F+SIG(1,K)*NPM(K)/(NPM(K)-1)
c          ENDDO
         RIJ = SQRT( RIJSQ )
         IF (RIJ .GT. EPSA(ISP(I),ISP(J))) THEN
            IF (DIM .EQ. 3) THEN
c               VIJ = -F*COU3*RCUT/SIGMA/2.
c               VIJ = VIJ + RCUT*/RIJ
               VIJ = SIG(ISP(I),ISP(J))*RCUT/RIJ
            ELSEIF (DIM .EQ. 2) THEN
c               VIJ = -F*(6.-PI+LOG(4.)-4.*LOG(SIGMA/RCUT))/4.
c               VIJ = VIJ - LOG(RIJ/RCUT)
               VIJ = -SIG(ISP(I),ISP(J))*LOG(RIJ/RCUT)
            ELSEIF (DIM .EQ. 1) THEN
c               VIJ = F*SIGMA/RCUT/4.
c               VIJ = VIJ - RIJ/RCUT
               VIJ = -SIG(ISP(I),ISP(J))*RIJ/RCUT
            ENDIF
         ELSE
            RIJ = EPSA(ISP(I),ISP(J))
            IF (DIM .EQ. 3) THEN
c               VIJ = -F*COU3*RCUT/SIGMA/2.
c               VIJ = VIJ + RCUT*/RIJ
               VIJ = SIG(ISP(I),ISP(J))*RCUT/RIJ
            ELSEIF (DIM .EQ. 2) THEN
c               VIJ = -F*(6.-PI+LOG(4.)-4.*LOG(SIGMA/RCUT))/4.
c               VIJ = VIJ - LOG(RIJ/RCUT)
               VIJ = -SIG(ISP(I),ISP(J))*LOG(RIJ/RCUT)
            ELSEIF (DIM .EQ. 1) THEN
c               VIJ = F*SIGMA/RCUT/4.
c               VIJ = VIJ - RIJ/RCUT
               VIJ = -SIG(ISP(I),ISP(J))*RIJ/RCUT
            ENDIF
         ENDIF
      ELSEIF (POTK .EQ. 'QHS') THEN
c          F=0.
c          DO K=1,NMIX
c             F = F+SIG(1,K)*NPM(K)/(NPM(K)-1)
c          ENDDO
         IF (DIM .EQ. 3) THEN
            IF (RIJSQ .LE. SIG(I,J)*SIG(I,J)) THEN
               VIJ = 10.**10.
            ELSE
c              VIJ = -F*QHS3*(SIGMA-SIG(I,J))/SIGMA**3/2./FTM(I)
            VIJ = VIJ + (2.*RIJSQ+SIG(I,J)**2.)/
     :                  (RIJSQ-SIG(I,J)**2.)**2./2./FTM(I)
            WIJ = -2.*RIJSQ*(RIJSQ+2.*SIG(I,J)**2.)/
     :                      (RIJSQ-SIG(I,J)**2.)**3./FTM(I)
            ENDIF
         ELSE
            WRITE(*,*) 'QHS in dimension different from 3'
            STOP
         ENDIF
      ELSEIF (POTK .EQ. 'HARMONIC') THEN
         VIJ = 0.5*RIJSQ/RCUT**2.
      ELSE
         WRITE(*,*) 'NO PAIR POTENTIAL AVAILABLE!'
         STOP
      ENDIF

      RETURN
      END


      SUBROUTINE GOFR ( C, DENS, MAXBIN, HIST11, HIST12, HIST22 )
      IMPLICIT NONE
C    ****************************************************************
C    ** CALCULATES THE RADIAL DISTRIBUTION FUNCTION HISTOGRAM.     **
C    ****************************************************************
      INCLUDE     'mc-mix.par'

      REAL*8        RXIJ, RIJSQ, RIJ, DENS
      INTEGER*8     C, IDIM, I, J, BIN, MAXBIN
      INTEGER*8     HIST11(MAXBIN), HIST12(MAXBIN), HIST22(MAXBIN)

      DO I = 1, NP - 1
         DO J = I + 1, NP
            RIJSQ = 0.
            DO IDIM = 1, DIM
               RXIJ  = RX(IDIM,C,I) - RX(IDIM,C,J)
               RXIJ  = RXIJ - ANINT ( RXIJ/SIGMA ) * SIGMA
               RIJSQ = RIJSQ + RXIJ * RXIJ
            ENDDO
C     CALCULATE HISTOGRAM FOR g(r)
            RIJ = SQRT ( RIJSQ )
            BIN = INT ( RIJ/DELR ) + 1

            IF (BIN .LE. MAXBIN) THEN
               IF (ISP(I) .EQ. 1 .AND. ISP(J) .EQ. 1) THEN
                  HIST11(BIN) = HIST11(BIN) + 2
               ELSEIF(ISP(I) .EQ. 2 .AND. ISP(J) .EQ. 2) THEN
                  HIST22(BIN) = HIST22(BIN) + 2
               ELSEIF(ISP(I) .NE. ISP(J)) THEN
                  HIST12(BIN) = HIST12(BIN) + 1
               ENDIF
            ENDIF

         ENDDO
      ENDDO

      RETURN
      END


CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC
c     determines the sign of the free spinless Fermi density matrix
c     dim  = spatial dimensions
c     nf   = number of fermions
c     beta = inverse temperature
c     itau0= initial time slice
c     itau1= final   time slice
c     sign = sign of the free fermion density matrix
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC
      subroutine ifdm(ispec,itau0,itau1,sign)
      implicit none
      include 'mc-mix.par'
      integer*8 ispec,itau0,itau1,i,j,ii,jj,idim,npp
      real*8 sum,DMGT,epsilon,det,sign
      real*8 aaa(np,np),beta,lambda,dx

      epsilon=1.d-6
      beta=ls*abs(itau1-itau0)
      lambda=1./2./ftm(ispec)
```

```
      if (itau0.eq.itau1) then
         sign=1.
         return
      endif

      npp=0
      do i=1,ispec-1
         npp=npp+npm(isp(i))
      enddo

c     set up the free particle density matrix
      do i=1,npm(ispec)
         do j=1,npm(ispec)
            ii=i+npp
            jj=j+npp
            sum=0.d0
            do idim=1,dim
               dx=rx(idim,itau0,jj)-rx(idim,itau1,ii)
               dx=dx-dnint(dx/sigma)*sigma
               sum=sum+dx**2.
            enddo
            sum=sum/(4*beta*lambda)
            aaa(i,j)=exp(-sum)
            aaa(i,j)=aaa(i,j)/(4*pi*beta*lambda)**(dim/2.)
         enddo
      enddo

      det=DMGT(epsilon,npm(ispec),aaa)

c     sign of the determinant
      sign=det

      return
      end


!The subroutine TSRGT applies to input real square matrix A(n,n) the upper
!triangularization algorithm of Gauss method with full pivoting and keeps
!trace of successive transformations done in integer vectors KP and LP.
!-----------------------------------------------------------------------------
!  Input parameters:
!  eps        precision (real*8)
!  n          size of A matrix (integer)
!  A          pointer to input real square matrix (real*8)
!  Output parameters:
!  it         flag=1 if A matrix ok, =0 if A matrix is singular (integer)
!  C          pointer to table storing main diagonal elements and supra-
!             diagonal elements of upper triangular matrix and the multi-
!             plying coefficients used during triangularization process
!  KP         table storing informations concerning the column exchanges
!             during process (integer)
!  LP         table storing informations concerning the line exchanges
!             during process (integer)
!-----------------------------------------------------------------------------
!The table C is first initialized to A matrix, then receives at each step k
!of the triangularization process, usefull elements of A matrix at step k for
!k=1,2,...n.
!The variables po(real*8), lo and ko(integer) store respectively pivot at step k,
!its line number and its column number.
!-----------------------------------------------------------------------------
      Subroutine TSRGT(eps, n, A, it, C, Kp, Lp)
      real*8 eps
      integer*8 n,it
      real*8 A(n,n), C(n,n)
      integer*8 Kp(n),Lp(n)
      real*8  po,t0
      C=A
      it=1
      k=1
      do while (it==1.and.k<n)
         po=C(k,k)
         lo=k
         ko=k
         do i=k, n
            do j=k, n
               if (dabs(C(i,j))>dabs(po)) then
                  po=C(i,j)
                  lo=i
                  ko=j
               end if
            end do
         end do
         Lp(k)=lo
         Kp(k)=ko
         if (dabs(po)<eps) then
            it=0
         else
            if (lo.ne.k) then
               do j=k, n
                  t0=C(k,j)
                  C(k,j)=C(lo,j)
                  C(lo,j)=t0
               end do
            end if
            if (ko.ne.k) then
               do i=1, n
                  t0=C(i,k)
                  C(i,k)=C(i,ko)
                  C(i,ko)=t0
               end do
            end if
            do i=k+1, n
               C(i,k)=C(i,k)/po
               do j=k+1, n
                  C(i,j)=C(i,j)-C(i,k)*C(k,j)
               end do
            end do
            k=k+1
         end if
      end do
      if (it==1.and.dabs(C(n,n))<eps)  it=0
      return
      End                       !TSRGT


!The function DMGT returns the determinant of a real square matrix
!A(n,n) by Gauss method with full pivoting.
!-----------------------------------------------------------------------------
!  Input parameters:
!  eps        precision (real*8)
!  n          size of A matrix (integer)
!  A          pointer to input real square matrix
!  Output parameters:
!  None
!-----------------------------------------------------------------------------
!The procedure TSRGT is used to reduce A matrix to an upper triangular matrix.
!Output variables are it(integer), C(n,n), Kp(n) and Lp(n).
!If it=0, matrix A is singular, if it=1, matrix A is regular. Table C contains
!at location i,j (j>=i) the corresponding element of the upper triangular matrix.
!Tables Lp and Kp contain informations relative to exchanges of line or column
!that occured during the process. For instance, the element number k of Lp is
!an integer <> k if an exchange of line has been made at step k (k=1,2,...,n).
!The number of exchanges of lines and columns is stored in integer L. the
!determinant of A matrix is stored in d0 (real*8).
!-----------------------------------------------------------------------------
      real*8 Function DMGT(eps, n, A)
      integer*8 n, it, Kp(n), Lp(n)
      real*8 eps, A(n,n), C(n,n)
      real*8 d0

      call TSRGT(eps,n,A,it,C,Kp,Lp) !call triangularization subroutine
      if (it==0) then
         d0=0.d0                !matrix singular, det=0
      else                      !matrix regular, det<>0
         d0=1.d0
         do k=1, n
   d0=d0*C(k,k)
       end do
       l=0
       do k=1, n-1
          if (Lp(k).ne.k)  l=l+1
          if (Kp(k).ne.k)  l=l+1
       end do
      if (MOD(l,2).ne.0) d0=-d0 !l is odd
      end if
      DMGT=d0                   !return determinant
      return
      End

*******************************************************************************
*** mc-mix.par ****************************************************************
*******************************************************************************

     REAL*8        PI
     REAL*8        DELR
        INTEGER*8     MDIM, N, MNP, MBIN, MMIX

        PARAMETER ( PI = 3.14159265358979323846264338328d0 )

     PARAMETER ( MDIM = 10   )   ! maximum number of dimensions
      PARAMETER ( MNP  = 1000 )   ! maximum number of particles
     PARAMETER ( N    = 3000 )   ! maximum number of time slices
     PARAMETER ( MBIN = 100000 ) ! maximum dimension of g(r) histogram
     PARAMETER ( MMIX = 2 )      ! maximum number of components
        PARAMETER ( DELR = 0.01d0 ) ! grid spacing for g(r)


     INTEGER*8     DIM, NMIX, ISP(MNP)
        REAL*8        RX(MDIM,0:N,MNP)
        REAL*8        FTM(MMIX), LS, SIGMA
        REAL*8        SIG(MMIX,MMIX),EPSA(MMIX,MMIX),EPSR,EPS,RCUT
        INTEGER*8     FTN0, NP, NPM(MMIX)
     INTEGER*8     NEXT(MNP), LEXT(MNP), MEXT(N,MNP), MBM

        ! path
        COMMON / BLOCK1 / RX, ISP, DIM, NMIX
        ! pair-potential parameters
        COMMON / BLOCK2 / SIG, EPSR, EPSA, EPS, RCUT
        ! mass, timestep, box edge, # timeslices, # particles
        COMMON / BLOCK3 / FTM, LS, SIGMA, FTN0, NP, NPM
        ! permutations
        COMMON / BLOCK4 / NEXT, LEXT, MEXT, MBM

*******************************************************************************
*** data-mix.in ***************************************************************
*******************************************************************************

0 spatial dimensions (1,2,3,...<= MDIM) nixture dimension (1,2,3,...<= MMIX)
3 2
1 seed of the random sequence                                        RAND
3
2 if bose                                                            (T/F)
T
3 number of particles NPM(I)                    (NP=NPM(1)+NPM(2)+...<= MNP)
10
10
4 the potential                                            SUBROUTINE POT
COULOMB
5 # time slices = 1/temperature/timestep                             (< N)
20
6 masses FTM(1), FTM(2), ...                                (hbar = kb = 1)
1.d0
1836.15d0
7 # of cycles                                                       (nstep)
500000000000000000
8 # of steps between output lines                                   (iprint)
100
9 # of steps between configuration saves                            (isave)
100
10 # of steps for equilibration                                      (iequi)
100
```

```
11 # of steps for acceptance ratios                    (iratio)
100
12 # of steps for rdf calculation                 (icalcg < #5)
10
13 configuration file name
conf.xyz
14 0 initialization, 1 restart, 2 restart rdf
0
15 density                                     THERMODYNAMICS
.16d0
16 temperature                                 THERMODYNAMICS
.005d0
17 maximum displacement/box edge
.01d0
18 maximum # of bridge timeslices                 (> 1; <= #5)
20
19 potential cutoff distance (LJ)              SUBROUTINE POT
1.d0
20 sig(i,j)                          (LJ 2.566)  SUBROUTINE POT
1.d0
-1.d0
-1.d0
1.d0
21 epsr                              (LJ INIT)   SUBROUTINE POT
1.d0
22 epsa(i,j)                                     SUBROUTINE POT
0.d0
.5d0
.5d0
1.d0
23 eps                               (LJ 10.22)  SUBROUTINE POT
1.d1
24 if only displace                                        (T/F)
F
25 if only bridge                                          (T/F)
F
```


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